

# Training: gbd (gradient-boosted trees)

HistGradientBoosting fitted on the tabular features, with early stopping on validation.

Gradient-boosted trees, fitted on CPU in about 2 minutes. They are an accuracy and feature-importance reference only: 500 trees are far too large for the microcontroller.

## How it is trained

- scikit-learn HistGradientBoostingRegressor: squared error, learning rate 0.05, at most 500 trees of depth 6 and 31 leaves, at least 100 windows per leaf, L2 1.0, random\_state 211.
- Data: all 20,755,925 training windows (features binned into histograms by the library), residual target dRR - rls\_pred in z units. Early stopping on the subject-disjoint validation windows (passed as X\_val), after 20 trees without improvement; it never triggered: both runs used all 500 trees.
- After fitting: permutation importance on up to 50,000 validation windows (5 repeats) and the Spearman correlation of all engine features on training windows, saved as feature\_importance.csv and feature\_correlation.csv in the run folder.
- Commands: python src/train.py --models gbd; python src/train.py --models gbd --features all --tag gbd\_all (run on CPU).

## Runs

| run     | features | trees | fit time | val RMSE (ms) | rls_ar (ms) |
|---------|----------|-------|----------|---------------|-------------|
| gbd     | 19       | 500   | 124.5 s  | 34.19         | 37.14       |
| gbd_all | 29       | 500   | 134.8 s  | 34.15         | 37.14       |

## What the training showed

- Both runs reached the 500-tree cap: more trees might still help, but the model is already about 238 KB at 8 B per node (budget 32 KB), so it was not pursued.
- gbd\_all (every engine feature) is only 0.04 ms better: its permutation importance ranks rls\_pred first by far (0.118 z), then rr\_dev, drr\_lag0..7, guzik and bp; the 8 RLS weights add nothing.

## Abbreviations and acronyms used in this document

| term        | meaning                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------|
| ar          | feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model    |
| B           | batch dimension (number of windows processed together) in the layer output shapes                 |
| bp          | feature: RSA band-pass output: RBJ biquad on raw RR, centre 0.30 cycles/beat, Q 1                 |
| CPU         | central processing unit                                                                           |
| dRR         | beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted |
| drr_lag0..7 | feature: drr_lag0..7: the 8 most recent dRR (lag0 = newest)                                       |
| gbd         | gradient-boosted decision trees (run name)                                                        |
| guzik       | feature: Guzik asymmetry: sum of positive dRR / sum of  dRR                                       |
| HRV         | heart rate variability                                                                            |
| KB          | kilobyte (1,024 bytes)                                                                            |
| L2          | L2 regularisation: a penalty on the sum of squared model weights                                  |
| ms          | milliseconds                                                                                      |
| RLS         | recursive least squares: an adaptive filter that updates its coefficients at every beat           |
| rls_ar      | baseline: adaptive per-patient AR(8) forecast by RLS                                              |
| rls_pred    | feature: RLS AR(8) forecast of the next dRR (ms), clamped to +/-300                               |
| RMSE        | root mean square error (ms): the main accuracy measure, lower is better                           |
| rr_dev      | feature: newest RR minus the window mean                                                          |
| val         | validation: the 16 validation subjects, used for model selection                                  |
| z           | z-score: (value - mean) / standard deviation, with statistics from the training subjects          |